



CERTIFICATE OF TESTS

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Printed: 05/16/2024

Heat: 2136635
Material: 5L-X52 4-1/2X.337 (15.00#)BR,DRL,PEB (11000102)
Vendor: Nucor Hickman - Armorel AR US (180436)

Purchase Order: 26632645
Sales Order: 1001000690
MFG Date: 03/15/2024

CHEMICAL PROPERTIES

	C	MN	P	S	NI	CR	CU	MO	SI	TI	V	CE
Ladle	0.0400	1.0800	0.0120	0.0013	0.0400	0.0500	0.1300	0.0100	0.2100	0.0140	0.0040	0.1118
	Al: 0.0300		Sn: 0.0050		N: 0.0060		B: 0.0000		Ca: 0.0020		Nb: 0.0330	
Chem	0.0447	1.2069	0.0112	0.0007	0.0383	0.0502	0.1304	0.0102	0.2402	0.0133	0.0043	0.1248
	Al: 0.0401		Sn: 0.0046		N: 0.0151		B: 0.0002		Ca: 0.0028		Nb: 0.0390	
Chem	0.0524	1.2118	0.0108	0.0003	0.0389	0.0506	0.1305	0.0102	0.2327	0.0137	0.0046	0.1327
	Al: 0.0408		Sn: 0.0048		N: 0.0156		B: 0.0002		Ca: 0.0043		Nb: 0.0406	

Acceptance Criteria: CEPCM = 0.2500

MECHANICAL PROPERTIES

YS (ksi): 70.18 TS (ksi): 75.78 EL (%) : 30.00 HRB: 87.00 Y/T(%): 0.93

CHARPY TEST RESULTS (BODY)

Orientation : TRANSVERSE	Specimen(S)	Energy (ft-lb)	%Shear
Spec. Size (in) : .394"X.197"X2.165"	1	75.00	100.00
Test Temp (°F) : -20	2	69.00	100.00
Location : 3T90	3	73.00	100.00
	Average	72.33	100.00

Acceptance Criteria Min. (ft-lb) : 10

COMMENTS

MINIMUM HYDROSTATIC TEST PRESSURE (PSI) : 4,700 PER 5 SECONDS

MANUFACTURED TO THE 46TH EDITION OF API 5L SPECIFICATION
MANUFACTURED IN THE USA-HIGH FREQUENCY WELDED PRODUCT
WELD SEAM ANNEALING AT 1600 DEGREES F.MINIMUM
ULTRASONIC TESTED TO N-10 NOTCH 1/8 DRILLED HOLE
FLATTENING TEST PASSED
1 INCH L90 STRIP SPECIMEN
THIS MATERIAL MEETS :ASTM A53B-22,ASME SA53B-21 LATEST EDITION
API 5L PSL 2 GRADE X52M
MEETS HARDNESS REQUIREMENTS OF NACE MRO 175/ISO 15156-2, CLAUSE A.2.1.1
MEETS ISO 10474 3.1.B/EN 10204: 3.1 / STANDARDS ARE LATEST REVISION
MELTED AND MANUFACTURED IN THE U.S.A.
STEEL SUPPLIER: Nucor Hickman - Armorel AR US
COIL MANUFACTURER: Nucor Hickman - Armorel AR US

The above analyses and tensile properties are correct to the best of my knowledge and belief.

Quality Control Dept